IMEN574 / IDSC574 · PROGRAMMING FOR AX

# 17

# Visual Representations

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Spring 2027

# Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

# Outline: Visual Representations

01 **Encoding Images**

From pixels to tokens

02 **Learning Representations**

From pairs to features

03 **Transferring Features**

From features to tasks

# 01 · Encoding Images


01 **Encoding Images**
From pixels to tokens

02 **Learning Representations**
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From features to tasks

# Why Representations

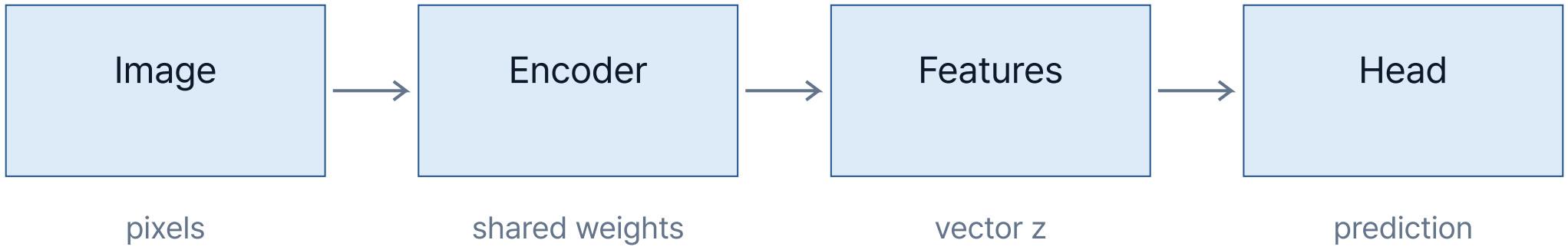
Pixel distance changes when the object barely changes

| | | | | |
|---|---|---|---|---|
| Image A | 0 | 0 | 1 | 0 |
| Shifted A | 0 | 1 | 0 | 0 |
| Background | 0 | 0 | 0 | 0 |

A one-pixel shift changes the array

# Representation and Readout

The encoder supplies features to a task-specific head




A new head can reuse the same features

# Global and Spatial Features

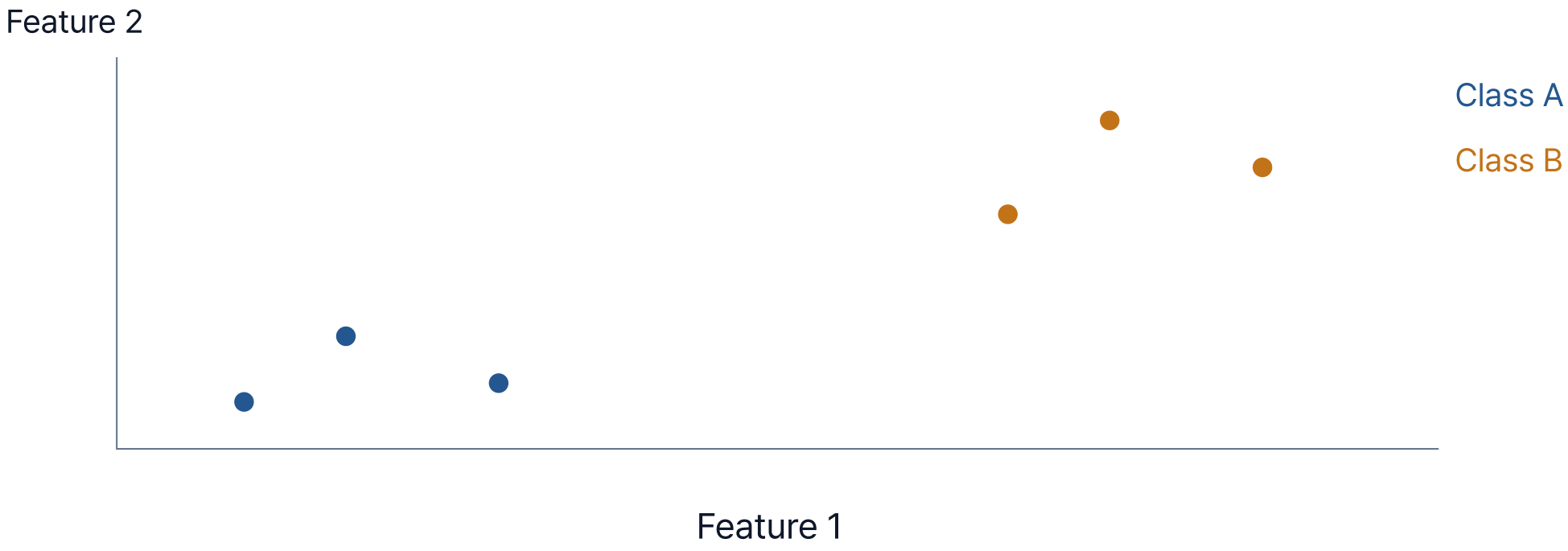
Pooling trades location for a compact summary

| | | | | |
|---|---|---|---|---|
| Patch 1 | 1 | 0 | 0 | 1 |
| Patch 2 | 0 | 1 | 1 | 0 |

Pooling removes the identity of each patch

# Supervised Visual Learning

Labels shape which differences matter




Nearby embeddings share the training label

# Local Receptive Fields

A convolution applies one local rule across an image

| | | | |
|---|---|---|---|
| top | 0 | 1 | 0 |
| middle | 1 | 4 | 1 |
| bottom | 0 | 1 | 0 |

The same kernel visits each local neighborhood

# Convolution Output Size

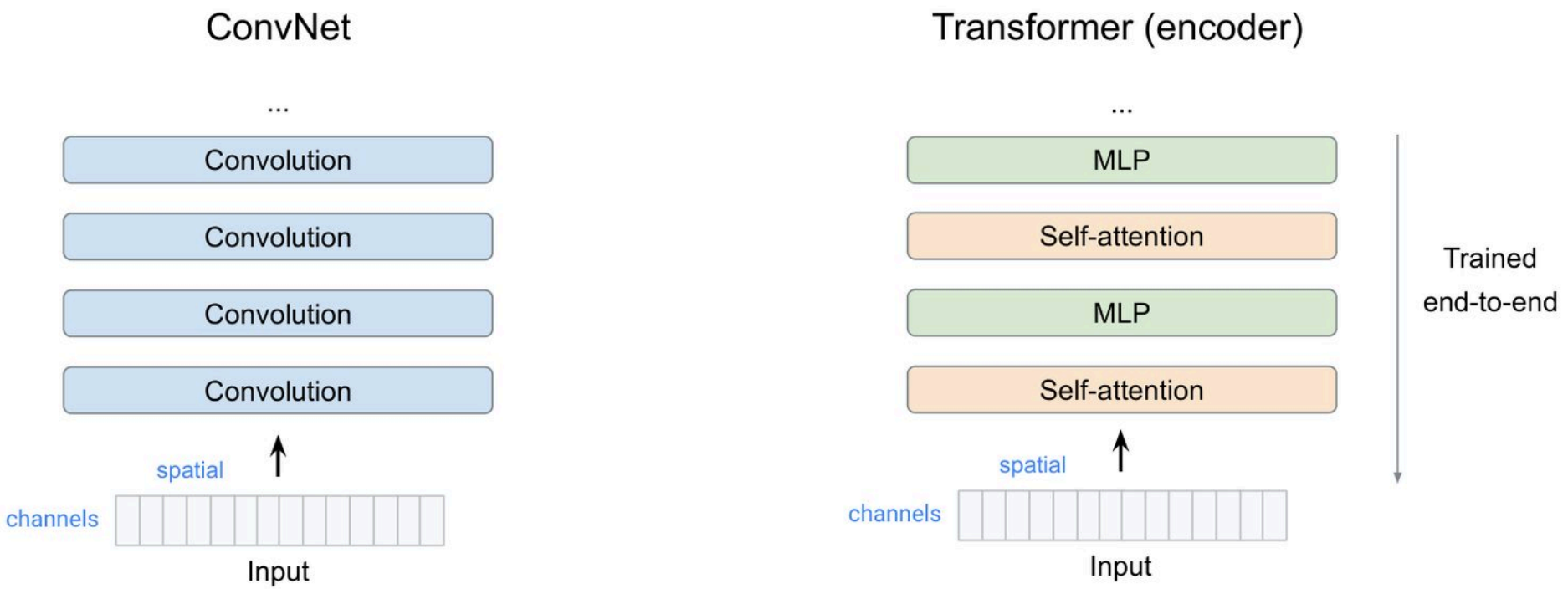
Stride and padding determine the spatial grid

$$H_{out} = \text{floor}((H + 2P - K) / S) + 1$$



# Convolution and Attention

Local filters and global token mixing impose different structures




Token mixing changes which locations can interact

https://arxiv.org/abs/2010.11929

# Patch Embeddings

An image becomes a sequence of patch vectors

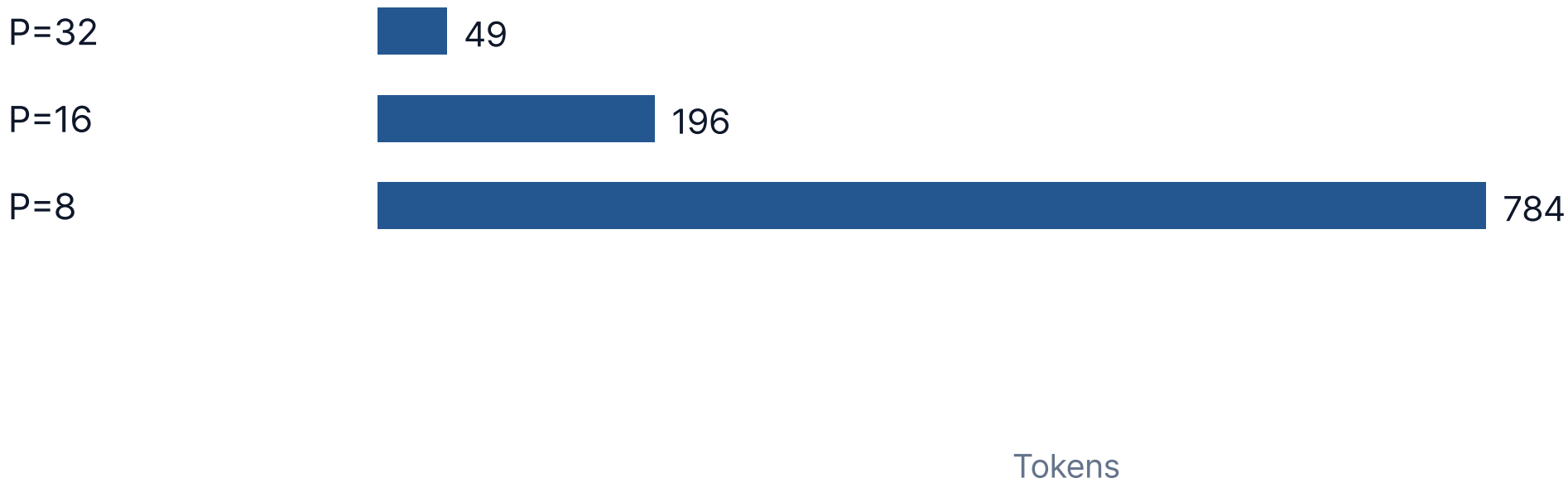
| | | | | |
|---|---|---|---|---|
| Image patches | p1 | p2 | p3 | p4 |
| Embeddings | z1 | z2 | z3 | z4 |

Each patch becomes one token

# Patch Count

Smaller patches create longer sequences

$$N = (H / P) \times (W / P)$$



# Worked Example: Patch Tokens

Trace the inputs before execution

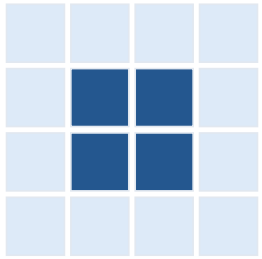
```
4×4 image; center 2×2 pixels = 1
Patch width = 2; horizontal shift = 0
Flatten each nonoverlapping patch in row-major order.
```

# Demo: Patch Tokens

Executable educational calculation; no pretrained inference

Change patch width from 2 to 4; compare token count and token width

Patch width 2 Horizontal shift 0



4 tokens × 4 values

Token 0: [0, 0, 0, 1]

Token 1: [0, 0, 1, 0]

Token 2: [0, 1, 0, 0]

Token 3: [1, 0, 0, 0]

Patch means

0.250 0.250 0.250 0.250

4×4 image → 4 tokens; token width=4.

Questions: Does shifting preserve tokens? Does mean pooling preserve position?

# Answer: Patch Tokens

Same inputs as the initial demo state

```
4 tokens, each with 4 values
each patch mean = .25
```

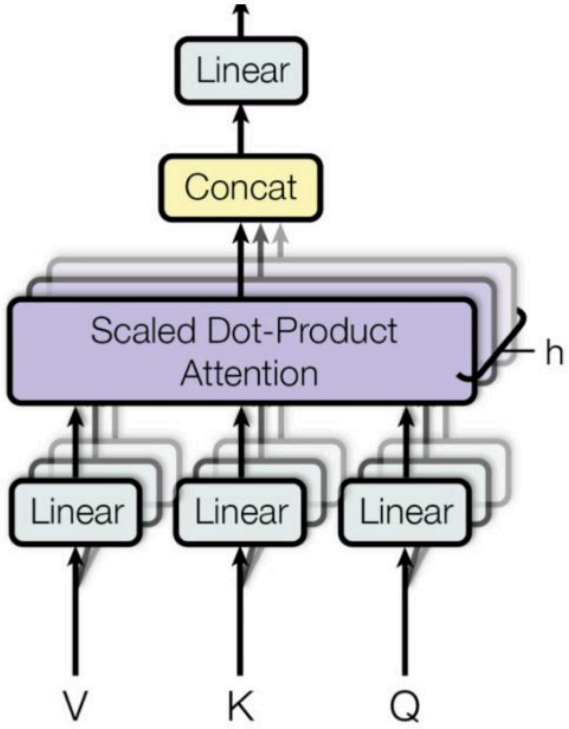
# Failure: Patch Tokens

Identify the assumption that breaks

```
Mean pooling discards within-patch arrangement
patch tokens are not learned semantic features
```

# Vision Transformer

Patch tokens enter a shared Transformer encoder




Patch tokens share the same encoder block

https://arxiv.org/abs/1706.03762

# Worked Example: CLS Attention Mask

Trace the inputs before execution

```
4×4 patch logits: central four=2, other twelve=0
Apply softmax across all 16 patches
Select weights ≥ .05; target mask is central four.
```

# Demo: CLS Attention Mask

Executable educational calculation; no pretrained inference

Raise the threshold from 0.05 to 0.20; compare mask size and target IoU

Attention threshold 0.05

CLS query scores → softmax across 16 patches

| | | | |
|---|---|---|---|
| 0.024 | 0.024 | 0.024 | 0.024 |
| 0.024 | 0.178 | 0.178 | 0.024 |
| 0.024 | 0.178 | 0.178 | 0.024 |
| 0.024 | 0.024 | 0.024 | 0.024 |

Selected patches=4; target IoU=1.000

Target = central four patches

Thresholded attention is a hypothesis, not ground truth

Fixed educational CLS logits: center=2, others=0; all weights computed by softmax.
Questions: Does a threshold change the learned features? Can an empty mask be a valid foreground?

# Answer: CLS Attention Mask

Same inputs as the initial demo state

```
Central weight .177809, background .024064
selected4 patches
IoU1
```

# Failure: CLS Attention Mask

Identify the assumption that breaks

```
Attention thresholding changes a proposed mask, not representation learning
this synthetic example does not establish SAM-quality segmentation
```

# Position Information

Permutation alone cannot recover the original layout

| | Left | Right |
|---|---|---|
| Top | top-left | top-right |
| Bottom | bottom-left | bottom-right |

Position identifies where each patch came from

# 02 · Learning Representations

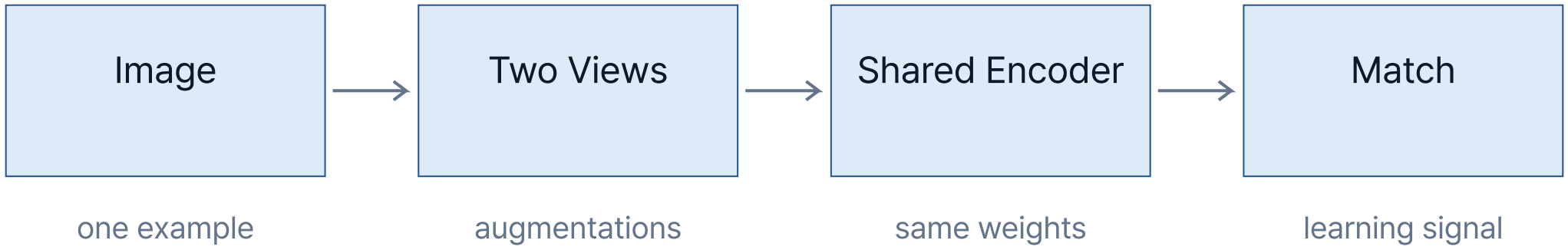
01 **Encoding Images**
From pixels to tokens

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From features to tasks

# Why Learn Without Labels

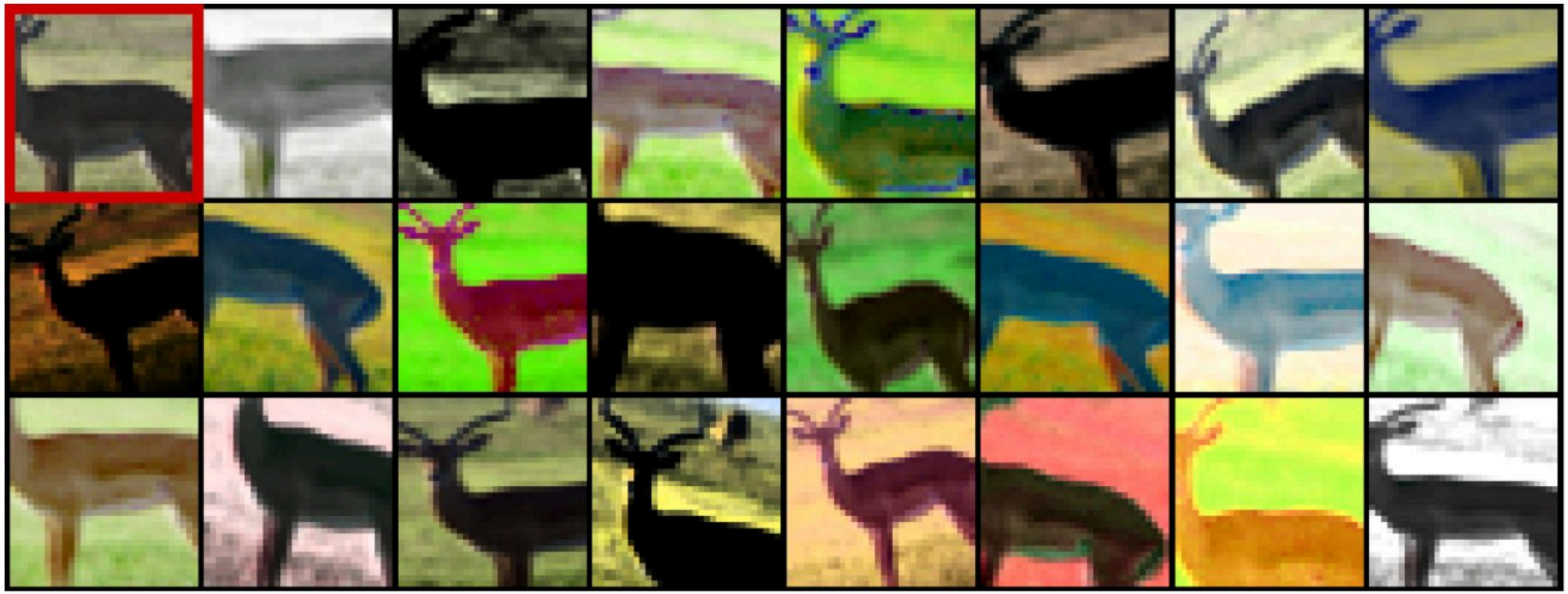
Unlabeled images still provide paired observations




Two views provide a learning relation

# Visual Cues

Different crops can preserve the same object



Different views can retain one object identity

# Positive and Negative Pairs

A training relation defines which features move together




Pair labels define the contrastive task

# Cosine Similarity

Normalization separates direction from vector magnitude

$$\cos(z, v) = z^{T}v \,/\, (||z||\ ||v||)$$

| | | |
|---|---|---|
| A | 1 | 0 |
| 2A | 2 | 0 |
| B | 0 | 1 |

# Similarity Matrix

Aligned views should score above mismatched pairs

| | view 1 | view 2 | view 3 |
|---|---|---|---|
| image 1 | 0.95 | 0.1 | 0.2 |
| image 2 | 0.15 | 0.9 | 0.05 |
| image 3 | 0.1 | 0.25 | 0.92 |

The diagonal marks matched observations

# Contrastive Objective

The matching example competes with alternatives

$$L_i = -\log[\exp(s_{ii} / \tau) / \Sigma_j \exp(s_{ij} / \tau)]$$

# Representation Collapse

Identical vectors satisfy matching without preserving information

| | | |
|---|---|---|
| Cat | 1 | 0 |
| Car | 1 | 0 |
| Tree | 1 | 0 |

Perfect agreement can erase every difference

# Avoiding Collapse

Additional constraints keep representations informative

| Family | Constraint |
| --- | --- |
| Contrastive | Compare other examples |
| Distillation | Teacher targets and asymmetry |
| Redundancy reduction | Control covariance |

# Worked Example: Paired-View Loss

Trace the inputs before execution

```
Views: [1,0], [0,1], [1,1] in both branches
L2-normalize; compare all 3×3 pairs; temperature = 1
Loss = −mean log softmax(similarity)[paired index]
```

# Demo: Paired-View Loss

Executable educational calculation; no pretrained inference

Compare Distinct with Collapsed features; explain why collapse has nonzero loss

Temperature 1 Features Distinct

Paired-image similarity

| | | |
|---|---|---|
| 1.000 | 0.000 | 0.707 |
| 0.000 | 1.000 | 0.707 |
| 0.707 | 0.707 | 1.000 |

Mean contrastive loss = 0.803

Positive P = 0.473, 0.473, 0.401

Diagonal entries identify paired views

Unit-vector dot products → row softmax → −mean log P(pair).
Questions: Can collapse give zero loss? What does temperature change?

# Answer: Paired-View Loss

Same inputs as the initial demo state

```
Mean contrastive loss = .803438
collapsed loss = log(3) = 1.098612
```

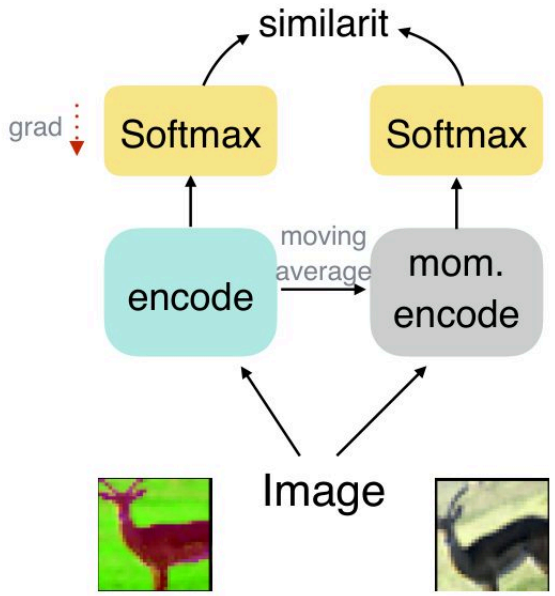
# Failure: Paired-View Loss

Identify the assumption that breaks

```
A low small-batch loss does not establish transfer quality
duplicated positives can become false negatives
```

# Teacher and Student

A teacher supplies targets for another augmented view




The target comes from another view of the image

# 03 · Transferring Features

01 Encoding Images

From pixels to tokens

02 Learning Representations

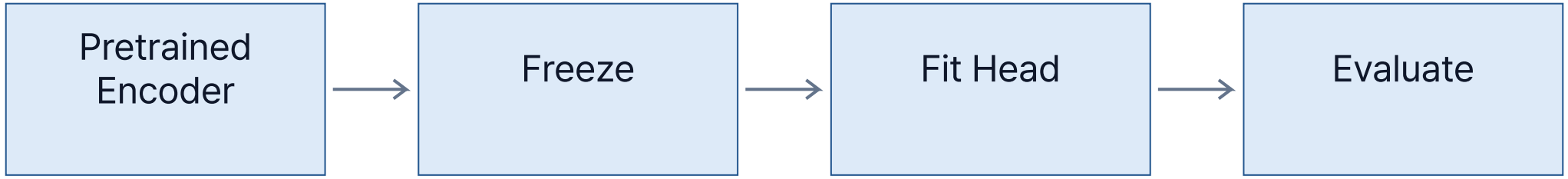
From pairs to features

03 **Transferring Features**

From features to tasks

# Frozen Feature Transfer

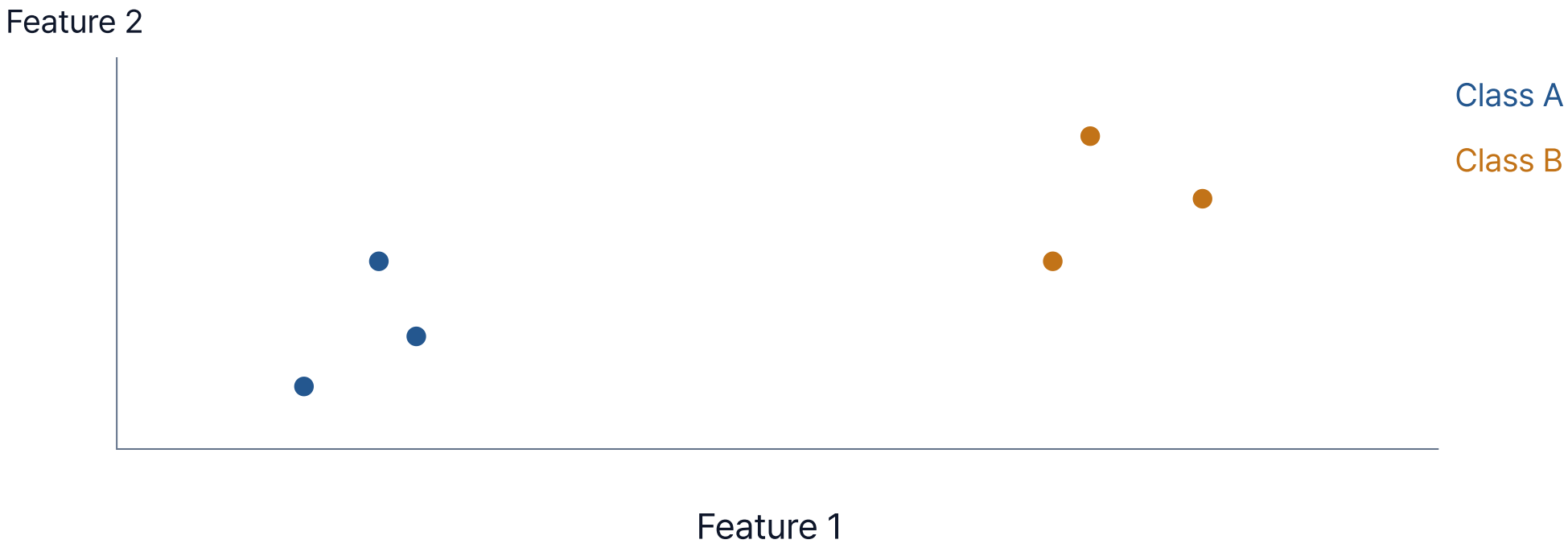
A small readout can reuse a pretrained encoder




Only the readout changes in a frozen probe

# Linear Probing

A linear head tests what is already accessible




A simple boundary exposes separable information

# Worked Example: Frozen-Feature Probe

Trace the inputs before execution

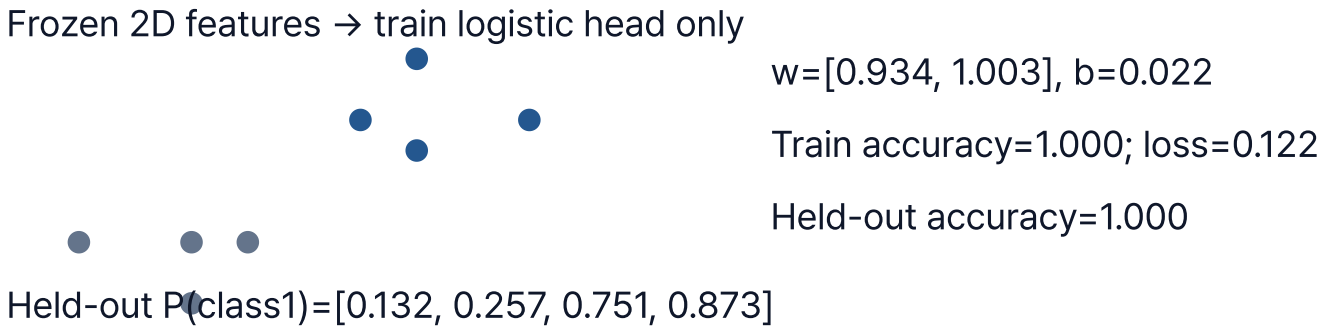
```
Fixed train features: 4 negative points, 4 positive points
Train logistic weights only; initialize w=[0,0], b=0
20 full-batch steps, learning rate .2; hold out 4 separate points.
```

# Demo: Frozen-Feature Probe

Executable educational calculation; no pretrained inference

Train the head for 20 steps; shift held-out x by 3 and compare both accuracies

Gradient steps 20 Learning rate 0.2 Held-out x shift 0




Full-batch gradient descent on 8 train rows; 4 held-out rows; no DINO checkpoint.
Questions: Which weights change? Why can held-out feature shift break a good probe?

# Answer: Frozen-Feature Probe

Same inputs as the initial demo state

```
w=[.933529,1.002926], b=.021973
training loss .121807
train accuracy1 and held-out accuracy1
```

# Failure: Frozen-Feature Probe

Identify the assumption that breaks

```
Frozen features do not mean the classifier is pretrained
toy probe accuracy does not measure DINO performance
```

# Fine-Tuning Scope

Updating more parameters increases adaptation and validation needs




Illustrative trainable parameter units

More trainable weights expand the adaptation space

# Augmentation Choice

Preserving the wrong invariant can remove the target

| | | |
|---|---|---|
| Color | red | green |
| Meaning | stop | go |

Removing color can remove the label

# Caveat: Background Shortcut

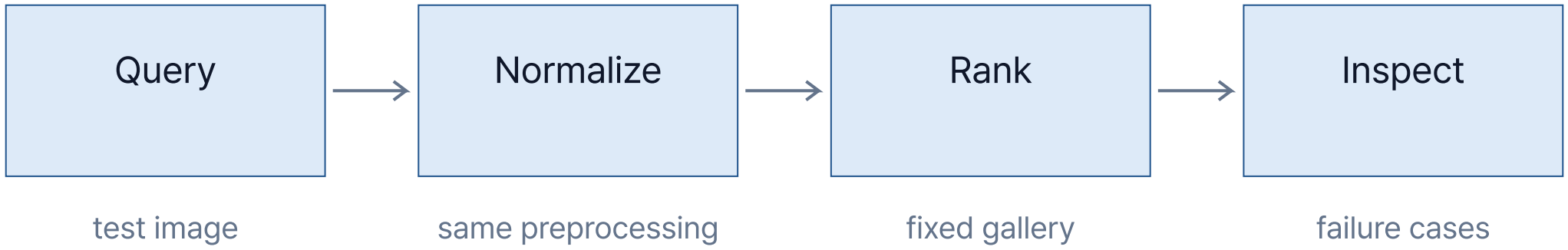
A classifier may learn context instead of the object

| | Class A | Class B |
|---|---|---|
| Train: snow | 8 | 0 |
| Train: grass | 0 | 8 |
| Test: mixed | 4 | 4 |

A background shortcut fails when context changes

# Retrieval Inspection

Nearest examples reveal useful and spurious similarities




The gallery determines what retrieval can find

# Pooling Code

Spatial tokens and global vectors have different shapes

```
tokens = np.arange(24).reshape(2, 3, 4)
global_vectors = tokens.mean(axis=1)
assert global_vectors.shape == (2, 4)
```

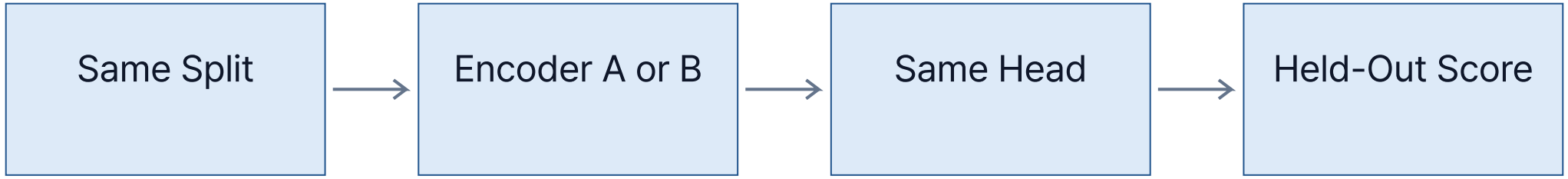
# Feature Normalization

Zero-length embeddings need an explicit policy

```
norm = np.linalg.norm(z, axis=-1, keepdims=True)
z_unit = z / np.maximum(norm, 1e-12)
```

# Transfer Evaluation

Hold the data split and readout fixed across encoders




A controlled comparison changes one component

# Representation Failure Audit

Check semantic, spatial and domain assumptions separately

| Question | Check |
| --- | --- |
| Meaning | Inspect nearest examples |
| Location | Keep patch tokens |
| Domain | Evaluate shifted data |

# Run the Original Model Lab

Actual pretrained execution remains a separate, explicit runtime path

| Original cells (0-based) | Model workflow | Observe |
|---|---|---|
| 5,7,9,14–19 | DINO + model transform → Oxford Pets → frozen features → linear probe | Inspect real held-out predictions; do not assume high accuracy. |
| 21–29 | Classwise cosine → pairs → t-SNE + image thumbnails | t-SNE is visualization, not held-out evidence. |
| 31–36 | CLS attention hooks → layer/head maps → threshold mask | A salient attention map is not a validated segmentation mask. |

Original model code remains available through the original-lab panel

# Takeaway: Visual Representations

01 Features determine what a small head can learn

02 Pair construction encodes an invariance assumption

03 Transfer requires a controlled held-out evaluation